

Advanced (Habanero)

- Q1.** What is the graph of the inequality $x + 2y > 4$?
 (A) Solid line with shading above the line
 (B) Dashed line with shading below the line
 (C) Solid line with shading below the line
 (D) Dashed line with shading above the line
 (E) No shading
- Q2.** Which inequality has a solid boundary line?
 (A) $x + y \geq 2$
 (B) $x + y > 2$
 (C) $x + y \leq 2$
 (D) $x + y < 2$
 (E) $x - y = 2$
- Q3.** What is the solution region for $2x - 3y < 6$?
 (A) Below the line $2x - 3y = 6$
 (B) Above the line $2x - 3y = 6$
 (C) On the line $2x - 3y = 6$
 (D) To the left of the line $2x - 3y = 6$
 (E) To the right of the line $2x - 3y = 6$
- Q4.** Which of the following inequalities has a dashed boundary line?
 (A) $x + y \geq 2$
 (B) $x + y > 2$
 (C) $x + y \leq 2$
 (D) $x + y < 2$
 (E) $x - y = 2$
- Q5.** The inequality $y > -2x + 1$ has a?
 (A) Solid line with shading above
 (B) Dashed line with shading below
 (C) Solid line with shading below
 (D) Dashed line with shading above
 (E) Vertical line with shading to the right
- Q6.** What type of line is used for the inequality $x - 2y \leq 3$?
 (A) Solid
 (B) Dashed
 (C) Vertical
 (D) Horizontal
 (E) Diagonal
- Q7.** The solution to $y < 2x - 1$ is?
 (A) Above the line $y = 2x - 1$
 (B) Below the line $y = 2x - 1$
 (C) On the line $y = 2x - 1$
 (D) To the left of the line $y = 2x - 1$
 (E) To the right of the line $y = 2x - 1$
- Q8.** The inequality $x > 0$ has a?
 (A) Vertical solid line with shading to the right
 (B) Vertical dashed line with shading to the left
 (C) Horizontal solid line with shading above
 (D) Horizontal dashed line with shading below
 (E) Diagonal line with shading above

Open Ended Questions (FRQ)

Q9. Graph the inequality $y > 2x - 3$ and identify the solution region.

rac12x - 3 and identify the solution region.

- Q10.** Write an inequality with a solid boundary line that has a solution region below the line $y = -x + 2$. Show the graph and explain your reasoning.

Chef's Tips

- Tip 1: Emphasize the importance of identifying the type of boundary line (solid or dashed).
- Tip 2: Make sure students understand how to determine the solution region based on the inequality.
- Tip 3: Encourage students to use graph paper to visualize the inequalities.